

ActInf GuestStream #028.1 ~ Reconceiving rationality: situating rationality

GuestStream | Reconceiving rationality: situating rationality into enactive cognition | 1:19:12

Hello and welcome everyone. This is Active Inference Guest Stream 28.1. It's October 14th, 2022. Today we are going to be hearing and discussing Reconceiving Rationality by Giovanni Rola. So Giovanni, thank you so much for joining. Very much looking forward to this presentation and discussion. So off to you. Thank you. Thank you, Daniel. It was a pleasure to receive your invitation a couple of months ago and I am delighted to be here to have this opportunity to discuss my work. As per your request, our suggestion, I am presenting this paper I wrote a couple of years ago and as we were talking the backstage, I have, I've changed my mind a few times and I'm going to make this clear during the presentation since I published that paper. But it is a pleasure nonetheless to have the chance to present this view here and to share it with the world. So I'm very happy. So we'll be talking about Reconceiving Rationality and Contentless Cognition. Then I'll present my view, which I called radically inactive rationality, and say a few words about agency and proficient interaction and reasoning and rationality. I'll close with a couple of objections. The first one being that whatever I'm presenting here is too liberal, in a way, excessively liberal. It encompasses too much. And the second objection, which I think is the one that may interest you the most is whether it is in fact compatible with predictive processing or predictive coding, which is of course in the vicinity. It's near the, of course, central to active inference. So here is the, here is the, sorry, the, my initial thoughts on the matter is that traditionally the very orthodox view on rationality assumes to the most of my knowledge, it's an assumption really that rationality must entail reasoning. That for instance, a belief B and an action A are rational if and only if there is the reasoning that underwrites the B or A. So, and of course, it's just not any reasoning. The reasoning in question must

taccord with certain certain formal benchmarks such as classical logic or probability reasoning or heuristics. So believe in

and acting rationally according to this traditional view require meeting certain formal standards. And this, as I said,

it seems to me to be largely an assumption about what it means to be rational about defining rationality in terms of reasoning.

So if rationality requires reasoning, and if reasoning involves the evaluation of beliefs against certain formal benchmarks, then rationality implies contentful cognition.

Where content, I take it to be minimally having accuracy conditions, being capable of being right or wrong, or in a more weightier notion being true or false.

So the idea here is that rationality necessarily involves content. It involves beliefs, reasoning, and the comparison or...

Sorry.

Sorry.

It's about to update right now.

It's about to update. Sorry, there was a pop-up in my window over here.

So that would be, I think, the standard view on rationality.

And if it is so that rationality necessarily involves content, it is a good fit with classical cognitivism or cognitivism in general,

which is the view that cognition necessarily involves representational content, at the very least.

For some cognitivists, of course, it might involve also conceptual content, but at the very least, representational content, which has accuracy conditions.

But this classical view is not so good, does not provide a so good fit with radical inactive and embodied cognition, which I am here thinking not specifically on radical inactivism by Vuto and Mien in the recent books.

But I think it also challenges other views that are not presented as radically inactive, but of course they share some core tenets with radical inactivism.

We would call them radically embodied views of cognitivism, such as the views presented by Ezequiel Di Paolo, Burman and Barania in *Sensory Mother Life*.

You also have Gallagher's variety of an activism in his 2017 book and also the view by Tony Kemerio in his 2009 book on ecological psychology and dynamical systems.

They all share common tenets and here I am.

I know this is very controversial, but I'm treating those views as radically embodied views of cognition and but I'm specifically focused on radically inactive views of cognition.

They all say that not all cognition is representational contentful, which is in a way to say that we can think of basic contentless forms of cognition or a contentless level basic level of cognition, which emerges through the active exploration of one's environment.

And beyond radical inactivism by Ruto and Mien, this idea could be cashed out in terms of mastering sensory modern engagements, a matter of know-how, knowing how to engage with your surroundings instead of knowing that stuff are so and so around you.

So this is very clearly what Di Paolo and colleagues and Kimero, for instance, do in their books radical activism has a certain tendency to skew sensory modern engagement sensory modern schemes, etc.

And but recent developments have, I think, incorporated that.

So this is the the the this idea, right?

So if we combine radically embodied views of cognition or radical inactivism and the classical view on rationality, what we have is something that Engel has described as follows.

Inactivism is the view that cognition is grounded in a prerational understanding of the world that is based on sensory modern acquisition of real life situation.

And he explains this.

this idea of this idea of prerational understanding by the ideas of Melopontine Heidegger, according to whom the relation to the world can be only one routine in practice, in acting and practicing in turn is mediated through the body.

So what seems to be implicit here is that unlike acting and practice rationality is not actually bona fide embodied.

It is something that is something that is not at the very basic level of cognition.

So this is what happened when we make room for a basic level of cognition rooted in act and in practice and divorce it from upper levels of cognition that are usually the locus, the loci class of rationality.

So the idea in summation is that contentless cognition or basic cognition that which emerges through sensory modern engagements is not rational.

So sensory modern engagements, however, virtues and prolific, they can be cannot be evaluated from the standpoint of rationality.

only contentful cognition, high level cognition can be rational, cognition that is expressed, for instance, in beliefs and reasoning.

And well, this is something that I haven't touched upon on the paper, but it was my implicit motivation to deal with this subject.

As someone that comes from philosophy of the mind is the idea of the myth of the given, at least a version of it.

It seemed to me at the time, and I was correctly persuaded by a reviewer not touching the subject.

It was very convoluted in my initial presentation, but it seems to me that if we divide basic cognition from rational thought and action, and we do not point towards a

the way that there is no intersection between these two levels, it seems that radical and activism and similar views would be committed to the myth of the given.

And popularized by sellers and more recently by McDowell as a problem for empiric views on the mind.

So it seems that this is, I think, a philosophically interesting problem, but it's not something that is on the forefront of the paper.

It was my implicit motivation here that to explain how can we combine the contentless or basic level of cognition through sensory motor engagements with rationality.

And if we cannot, it seems that there is a given of some sort that is being not completely explained in our picture of the mind.

So this is something I wanted to do. I still want to try to explore this, this intuition more clearly, but I haven't touched upon this.

But I think there's something in the direction of what I will show today is, I think it's a way to ease this accusation of being committed to the myth of the given.

So how can we reconceive rationality? I mean, I think in order to avoid that issue and avoid the idea that we have two levels completely separated from each other,

where one, where one, the higher one is the locus, classical rationality and what else is benefit is not rational.

I think the idea is to reject that rationality must be defined or entails necessarily reasoning.

Rationality does not need to be so conceived, but then the challenge becomes to offer a positive proposal of what rationality is, if not reasoning.

And here is Susan Hurley in a paper that to my knowledge is the only one that touched upon this subject.

And I think she was also concerned about the myth of the given, by the way, at the time.

And here is a long quote from here, if you know, might be reading it is that rationality might emerge from a complex system of the centralized higher order relations of inhibition,

facilitation and coordination among different horizontal layers, each of which is dynamic and environmentally situated.

Rationality reconceived in horizontal or modular terms.

So in terms, I don't know if she has to use this terminology, but anyway, that's how she chose to express it, is substantively related to the environment.

It does not depend only on internal procedures that mediate between input and output, such as content, either for the organism as a whole or for a vertically bounded central cognitive module.

Rather, it depends on complex relationships between dedicated world involving layers that monitor and respond to specific aspects of the natural and social environment and of the neural network and register feedback from their responses.

So this is, to my knowledge is the earliest case of someone working on the inactive approach in a way.

Her play is obviously more usually associated with the work of Alva Noh, which is also an activist, but in a different approach from radical inactivism.

And here is the earliest and then to my knowledge before I wrote that paper is that the clearest proposal of rationality reconceived in the inactive terms, but it's not something that gained traction after publication.

To my knowledge, at least I haven't seen much since then.

And to summarize the idea is that we have to change the explanatory focus, which is rationality is not exclusively thought of as a content involving capacity in this proposal.

It is situated or world involving in early words.

And is it is radically embodied.

It depends on bodily morphology and action and how the organism is situated and acts in that environment.

And the idea would be minimally that.

So this is what I call radically inactive rationality.

Is in a very loose definition so far is that an Asian acts rationally insofar as she maintains a proficient interaction with her environment, coordinating her abilities according to environmental conditions of constraints and solicitations.

And of course, this sometimes requires adapting to unforeseen circumstances.

The core intuition is this.

And it's interesting to compare this idea of radical inactive rationality with another of Susan Hurley's ideas in a text from 2003.

She says reasons for acting that she argues that reasons for acting may be context bound where she's arguing against Gary Tevland's generality constraint.

And this is the defense of her idea of islands of practical rationality.

So she is still treating rationality as essentially something that relates to reasoning here.

But this kind of reasoning is bounded by practical context.

So this is a very progressive view if compared with the traditional view I talked about at the beginning, but it's still more conservative than what I am trying to develop here.

So another thing that I think must be in the forefront of this proposal is that rationality implies agency and autonomy because something or only something that an organism does can be said to be rational.

Right?

So to use these ideas from the autonomous and activist camp and from the theory of autopayesis, a system such as a rumba cannot be rational because it's not an agent.

It is not the author of its own behavior.

It is designed by external agents to work in a certain way.

So it's not strictly speaking an agent.

So that is where I would draw the line of where, well, you cannot, it makes no sense to ascribe rationality to a rumba for instance.

And for this, I, I use the notions of autonomy and adaptivity.

The DiPaolo and others, you have also the, the recent book I mentioned before, Sensory Mother Life and that famous paper from DiPaolo from 2005, where he talks about the adaptivity to, to set his very general constraints on what it means to be rational.

So autonomous systems are thereby conceived as precarious networks, a precarious network of processes because organization and physical boundaries are produced by the system itself.

The system is continually maintaining its own viability conditions, which is known as sense-making where the system explore environmental, environmental conditions that are favorable to its, uh,

self-maintenance and avoids, uh, self-maintenance and avoids the ones that are detrimental to it.

So this is how I understand to be that the, the very, the, in general, that the, the most general constraint on a rational being that it must be an autonomous system.

And the interesting thing here is that the concept of adaptivity that DiPaolo puts forth is that the capacity of an organism to improve its condition of self-production through internal structural changes.

So it involves the monitoring and regulation of internal changes in order to select the best available course of action.

This, I think it's another interesting concept to think of, uh, to, to, to include in this idea of a radically inactive notion of rationality, one that is not necessarily committed to reasoning and content.

So agency is realized through the exercise of situated abilities that promote changes within the organism in accordance with environmental constraints and solicitations.

Which means that how well an organism adapts in order to deal proficiently with its environment is, um, a measure of, or an indication of its rationality.

The more rational an organism is, the more suited it is to deal with its environment and with and for seen circumstances.

And in doing so, the organism enhances the continuous exchanges between, uh, itself and the environment, which is crucial for its survival.

And in this very naturalized way of thinking of rationality is mostly capacity to, to enhance chances for survival.

This is the, um, the overall idea here.

So as I said before, what about reasoning?

I want to, I don't want to commit rationality necessarily with reasoning.

And this is what radically inactive rationality is all about, but I don't want to exclude reasoning as a form of rational cognition.

So what I'm saying is that it's not defined by reasoning, but it's not, uh, it does not exclude reasoning.

Reasoning, I think would be thought of as a special or a specific mode of rational agency.

One that is exclusive to our knowledge to human beings, but it's not the defining feature of rationality.

So for this, I, uh, in order to, to explain this idea, I recall how Hutu and Satine and Hutu and Min, in the, uh, in a paper in the most recent book by Hutu and Min,

they explain the emergence of higher cognition.

The idea is that being able to engage in contentful cognition ultimately depends on being socioculturally embedded,

where contentful cognition emerged from sociocultural interaction.

So as an individual takes part in sociocultural practices, they, uh, accept more ancient cognitive abilities to deal with symbols.

And by doing so, they, um, in a long process of, uh, inculturation, of course, they become able to engage with, uh, symbolic cognition,

which is not internal representations per se because it is publicly available.

It's not mental representations in the traditional sense, but it has the features of symbolic cognition without the, the commitment to internal representation or mental representations.

So acting rationally in its most general sense is the idea that the proficient behavior of an organism is, uh, is, is, is the proficient behavior of an organism in its environment

and depends on the coordination of embodied abilities.

Whereas reasoning, if everything goes well, of course, is understood as contentful rationality, form a specific mode of rationality.

Reasoning is the proficient behavior of an organism in a sociocultural environment.

which is a more specific aspect of our environment, of course.

And it depends on the coordination of sociocultural based or content involving abilities, which are more recent in the, the human phylogenetic scale.

So, well, this is the, the biggest of the idea.

This is the, the, I think, uh, the, the propositive part.

And here are the objections I wanted to talk about.

First one being that it's a very liberal view.

It's very, uh, it, it, it is, well, it encompasses too much one would say, because if rational action boils down to the coordination of cognitive abilities, then rationality is something gradual for an organism could be more or less proficient in its environmental interactions.

And the, the, the objection goes according to radically inactive rationality, many organisms achieve and maintain an adaptive behavior in the niche with different levels of success.

To which I am, uh, well, someone could say rational and active, uh, radical and inactive rationality is too liberal.

And this is one objection that I I've heard in person and I, I kept it to heart because I think it's very, it's a very good way of putting it.

Even snails are rational in this view.

And, uh, to which I must say that I am willing to, to bite the bullet.

Uh, well, yes, it's more liberal than traditional views.

Of course, this intended to be so, but I don't think that this per se problem.

What I think is the, the, the positive part here is that it preserves a difference of degree.

Humans are way more rational than other animals, simply because we are able to perform a wide variety of actions in a wide variety of dynamic circumstances that are way beyond any other animals that we are aware of.

So to come back to the example of this nail, this nail can perform a few complex actions.

It can move from one point to another.

It can search for food and shelter and so on and so forth, but they have a critically limited set of abilities.

If you surround this nail with a salt, for instance, it is doomed to, to die by the irritation.

And humans, on the other hand, are much more adaptive and able to deal dynamically with many unforeseen things.

circumstances.

So, uh, my response to that objection is that yes, uh, well, it is very liberal, but that's not the problem.

The problem would be if that it did, if it had the room for a difference of degree between how animals deal with the environments.

So, and, and this is the other object objection, which I think it may be the, the one that interests you the most.

Then, you know, is that well, predictive processing seems to imply inferences and content all the way.

Uh, but at first I must, uh, warn you and the viewers that at the time I wrote this, uh, this paper, I was much more optimistic about this possibility of combining an activism and predictive coding or predictive processing.

And, uh, uh, I think there are important points here, but there is this recent paper by, uh, and, uh, Evan Thompson and Randy beer, where they, uh, they provide a critical assessment of, uh, the, the, the, the attempts to combine mainly the free energy principle.

Uh, uh, uh, uh, uh, inactive and, uh, autopoietic theories.

So I am now mostly convinced by their arguments, but I think there are some interesting points of comparison here nonetheless.

So predictive processing or predictive coding, uh, at the time I was writing in is still to this day is all the rage in the cognitive sciences.

I guess you can say that it's on its way to become one of the main paradigms out there.

So it is interesting to see if it provides a good fit with what I'm talking about.

And in this view was, you know, cognition is understood as a process of error minimization that occurs in bidirectional neural hierarchies.

So you have the generation of internal models for the activity of lower layers that anticipates the distal causes of sensorial stimuli, which is a top down process and the minimization of discrepancy between prediction and current stimulus signal by, um,

uh, uh, the optation of priors, which is a bottom up process.

So here's, of course, active inference, which is understood by, for instance, in a, informal, in informal terms by Clark,

that an organism moves its sensors in ways that amount to actively seeking or generation, generating the sensory consequences that they, or rather their brains expect.

So this is one way of understanding active inference in very informal terms, but I think it gets the job done.

So is the brain inferring the expected source of received the stimuli?

If it is, it seems that there is a, there's an issue with what I'm talking about.

So the reading of predictive processing, which is, uh, which uses the product by as an inference, right?

Says, for instance, that the process of perception is tuss inseparable from rational broader by Asians process of believe fixation and contact stop-down effects are felt at every intermediate level of process.

level of processing this has called my attention the way that clark is also talking about rationality all the way but it's an influential sort of rationality and pervasive inference reasoning seems to imply pervasive content and then predictive processing seems to be incomparable with radically inactive rationality well but there is at the time there was and i think we can still argue in favor of the there is no known hemholzen reading of predictive processing which uses for instance one first step would be to use a deflationary sense of prediction that under saw in cameo in a short paper from 2013 but forth where the values of two or more variables co-vary or are very reliable so the value of one allows the observer to infer the value of another and here the inference is happening happening at the level of the observation not that one file of uh variable is inferring the value of another so the idea would be i think that content is in the eye of the beholder the person who interprets how the selection of sensorial input approximates internal predictions here understood in the deflationary sense can describe it in inferential and by easy terms so this is how i see for instance the the idea put forth also by bruneberg rietfeld and kiverstein in that paper which if i'm not mistaken title is something like the brain is not a scientist and the argument is in this general direction that we are projecting how scientists construed brain activity as a feature of the brain itself as if it were predicting and inferring this more robust or weightier sense so what i'm trying to say here is that active inference then i use double i'm going to say double i forgot the name of this in english sorry um it's not uh the literal sense of inference here according to radically inactive uh rationality is that a norm for the efficiency of goal oriented behavior the agent has to act in certain ways in order for his his or her states come to fruition more efficiently less costly this does not imply the more contentious claim that the generative models themselves display and perform predictive inferences this is a very deflationary reading into predictive processing to to provide this uh combination with radically inactive rationality and here is uh kickoff in a paper from 2018.

by model it doesn't follow that an organism has an internal representational model of its niche and that it is this model that does all the cognitive work if you like instead an organism is a model that is the causal and statistical regularities reflect in some environment are reflecting some phenotype that is a model and well this relates to something that recently uh my colleague nada figure and i have been talking about the idea of how organisms bring forth a world according to an activism which is idea that the organism and environment are historically historically codetermined so at a phylogenetical scale this could be explained through the idea of niche construction the idea that once an organism is active in its environment it promotes changes that if they are stable enough they can be inherited by its offspring and these changes provide new evolutionary press pressure pressures that are that lead its evolutionary pathway a similar idea has also been explored by evidence this time relating niche construction and ecological psychology so my my my response to that objection is that to act

rationality means to achieve a proficient engagement with the environment through in a way prediction and error minimization i'm willing to accept that characterization but rational and broadly by easy processes need not imply contentful cognition if we if we understand that in the flashner sense prediction and error minimization involve the selection of input and the coordination of linked abilities in world involving manners which is well exactly the main claim of radical and active rationality so i think to that extent that predictive processing is not uh an enemy would be rather an ally of this idea instead of uh uh a rival but of course there are other issues of compatibility that i haven't i haven't touched upon here but that uh is all that i have thought of for today thank you awesome thank you very much all right

cool very interesting for those who are watching live please feel free to write any questions i will gather my notes and let's begin the discussion okay well first just oh thanks again for the talk how about a little context on how you came to be asking these questions in your current position and how your interests in this area developed because there's so many interesting ideas and i just wanted to learn more about how you came to this framing and sharpened your point okay thank you that's that's a good question because it's not often that you can explore this background and matters so uh i was at time i think i was writing my phd dissertation it was maybe 2017 or something and uh i was arguing for uh in the idea that rationality must be something that is ultimately world involving there is a tendency to think of well even among as i told as i showed you among uh in in embodied periods with people of certain abilities to be essentially word involving such as acting and practice and so forth and perception of course but rationality remains something like a naturalizable notion something that cannot be uh brought down to the level of the could not be scaled down in a way unlike perception that could be explained by a certain dynamic engagement so on so forth so at the time i was thinking that well maybe this is an issue because on other uh uh other contexts it is taken to imply that whatever perception is if it is not articulated in a way to provide a certain fit with uh rational cognition then perception is a given and this is the myth of the given i mentioned at the beginning and i thought that well maybe this is something of there is something of a myth of a myth of the given working here on an active cognition but this is not i think it's not fatal because we could in principle argue that instead of uh construing the set perception and other basic forms of cognition as involving concepts and the like maybe we could argue from the other direction try to scale down rationality at a more naturalized level of uh perceptual cognition and the like so that was the motivation and uh it was very it was a long review process and i believe that my first the first uh version of the paper that the the reviewer was very i think they were correct entirely i i endorsed the suggestion after all but he was very um they were very adamant on taking that part and i think it was for the better because it it is another literature it is another uh a lot of reference that they are not usually in conversation with an activism and embodied combination and it was very difficult just to put the matter in the same terms to advance my argument so it's something that i scratched out and i haven't thought too much about that since then but i think it's something i would like to come back to and maybe uh even

probably someone has done so in the in this last few years i'm not aware i can remember but this relation between an activism the myth of the given it seemed to me it seems to be very obvious so it is i think something that we could we should i mean uh be able to explore but there's not something that was in the forefront of the paper but but i had this idea to talk about rationality in the flash in the flash in every sense and at the same time i was reading a lot about uh coding from a philosophical perspective of course i'm not as i said very uh well versed in the technicalities and the formalities of the the approach but i was interested in the philosophical consequences there and i thought well maybe there's something to compare here because as clark's uh quotations show you can talk about rationality all the way and uh that's something i wanted to to develop but not in a cognitivist view but not radically embodied film so that's the the the background awesome thank you could you maybe describe just what is the myth of the given what was the context that that myth arose in and where do we go after bringing up the myth of the given because i i saw it as a concept that you wove in so what was it about okay so the middle of the giving is that well there are several possible givens depending on your flavor of philosophy of mind but most classically empiricist views of the mind take the the the content of perception to be given prayer rationally in a way that we have certain inputs that are uh just received by the mind and the mind interprets it so sellers uh wilfrid sellers is famous for uh having described this uh this problem as a problem for not only implicit views of the mind but also for rationalists but on the on the other side rationalists usually take uh rational intuition as a given which is to say that there is some some level of cognition that is uh unarticulated it's given or some people talk about being pre-given but in a way there is this level of uh cognition that is given and the mind has to interpret it but if it is given in this way it has no point of contact no uh intersection between whatever whatever mind does and in this basic level of cognition so my idea is to avoid the the charge of being committed to the myth of the given but again this is not something that is explicit in the inactivist literature but it's something that seems to be possible to to develop in uh in uh in this sort of comparison but it's not it's not on the it's not clearly there but

uh one way of solving the myth myth of the given is to say that perceptual content is already conceptually nature which is for instance how john mcdowell does in mind and world that you know a perception uh the the joke is comes with uh subtitles you and interpret the world as as you do but uh uh i think that one way another way is to demitify in a way rationality and the upper layers of cognition try and try to think of them uh in less uh contentful in a less uh without committing them so strictly speaking with a level of content that is divorced from basic levels of cognition that's one way i think of doing so and rationality seems to be the one key concept to do this because it's so usually assumed to be something exclusively human and something that is related to reasoning and well if it is so then rationality cannot inform action and if of course if action is contentless in nature so if rationality is essentially essentially contentful and action is essentially uncontentful or contentless then you can you do not have this uh that's time this uh intersection between two domains and this is something i wanted to avoid but as i told you it became so hard to express these views in

in sufficiently clear and general ways that by recommendation of a reviewer i expressed it out and i i i still think about it but not in i haven't developed a proper line of argumentation that deals with it more clearly than i can do as i just did in very general terms

in very general terms cool well some words that you associated with rationality were adaptivity and also proficiency so what do you see as the relationship of rationality and fitness

fitness as a biological function right yes yeah okay i um i think it is it is an intrinsic relation really i think it is uh probably the most um the obvious consequence of rational action at the species scale when we talk about whether rationality is a feature of uh certain species or instead of individuals i think fitness is one of the the key concepts here to to think this as the the outcome of this rational action so although i do not say i do not put it in those matters in the paper i think it is possible to think it that way i don't see it as an uh as an objection to think think of rationality this way but again it's very liberal and i'm willing to to accept this consequence as well every living organism in a way if it if it hasn't gone extinct yet so it is in a way fitted to its environment so i'm willing to say that it may involve a minimal level of rationality of dealing with uh

a dynamically changing environment but well it's the price of having a very uh liberal progressive notion of rationalities this is a consequence of it i believe so you've said that fitness is is a consequence of rational behavior are there other consequences or how do we recognize rationality from the outside other than just saying well the ant colony is thriving so to the extent that it is rational it is adaptive and exhibiting agency and all of that um is this such a deflationary rationality concept that all extant things humans and otherwise are merely described by it or are there some outcomes that we can use to disentangle like for example irrational or irrational behavior from merely the consequences reflected by survival and fitness

right well i i have to to think very carefully about this because i don't want to trivialize this concept in a way that well everything is ipso facto rational by the matter of fact it is living it is acting rationally but i think that uh in particular cases one could uh identify the connection that are inimical to the individuals um self-sustainment in that niche so at the individual level in particular cases one can clearly see certain actions that could classify as irrational to the extent that they uh are detrimental to to its survival but uh at the species level it's more difficult to argue in that way because to identify this kind of behavior or consequence but i think that in order to not trivialize the idea of rationality and say well everything that is living is in fact you'd have to to identify this sort of particular acts that uh are detrimental to to one's survival so in the case of the the ant colony i don't know maybe um welcoming invasors would be uh would be the the sort of behavior that would be prejudicial to with uh survival as such i don't know i don't know if this is a it's one impossible way of thinking about this i had to think more clearly but okay well very interesting and you brought in in that answer and in the presentation like multiple nested time scales just like eco evo devo does from a more biological perspective looking at how niche construction bridges the gap between slower phylogenetic processes and more rapid developmental processes um so how does this

embedded and multi-scale consideration relate to the the content concept like is content going to be localized in one body or is the content distributed across entities or is the rationality distributed across entities how would we even identify the locus of rationality and that also kind of invokes the um horizontal horizontal and modular conception from the quote that you provided so how do we identify uh salient spatial temporal scales and systems of interest

well i i don't know i don't know if i can answer uh this fully but it seems to me that uh the level of uh if we're talking about content for instance content as a accuracy conditions for symbolic thoughts i think fortunately we have to say that it's it's not localized in a single individual it's publicly shared so you can say that um one consequence of this is that it's possible to think of different uh patterns of reasoning that have been uh been uh entrenched in different uh societies at different times etc and you can see that well they develop different level different uh rules for content for the nation and i think this is plausible to say that content is is localized that the the the uh how can you say is shared among you know the the core specifics of if you're talking about content for the nation at least but as for rationality i think one could um identify in particular behaviors if it it if it fits these uh this kind of description of uh being proficient a proficient interaction and adaptive interaction between an organism and uh its environment its niche so probably we could uh we could give different answers here for rationality and content i i am willing to say that contentful cognition is not something that happens in the head something that is out there in a way shared among others and this is something that the philosophy of mind has uh at least since the 60s uh the philosophers of mind have been argued for the idea that uh content is externally individuated and uh i'm i think this is it is one of uh the idea by for instance hillary putnam and tyler burge and others that mental content is individuated at the community level or societal level but on the other hand i think rationality could be in principle individuated in particular behaviors but it would depend on the individual's goals and the the niche it inhabits and so on so forth there are of course actual variables here that might be important to to identify behavior as rational or not

thanks um well one phrase that really stuck with me was like know how not know that perhaps uh a basal point in your work and a focus on the action so i just wanted to hear about how action and practice as you wove them here today how are those terms addressed and how are things different today with the treatment of action and practice and know how not know that than say in the helmholtzian days or even in ancient or in world knowledge traditions so okay cool so the idea i think would be that um there is a tendency from uh it is a tendency from a very least modern philosophy but it has been very influential in the first wave of cognitive science to think of perception as a capacity to categorize the environment and to offer a discrete description of environmental aspects so for instance perceiving that uh the the the the i don't know the the curve is to my left is a matter of knowing that there is a thing with such and such properties to my left and this is a kind of uh know that a kind of propositional attitude towards the world so uh i think the the the interesting thing about an activism and ecological psychology as well and to the extent that they recall the american pragmatism from the the beginning of the century is to think of perception as

a matter of not categorizing discrete properties in the world but as uh disclosing possibilities for acting and this is something that if we were to explain that in terms of uh in epistemic terms such as knowledge we wouldn't put it as a matter of knowing that the the the cup is to my left but knowing how to grab the cup and do whatever i want to do with it if i want to drink a cup of coffee or if i want to use it as a paperweight it will have different silent properties and this is something that although it has been i think very i think an activism in ecological psychology and other embodied views of cognition have been largely successfully in uh in the uh in the uh in the uh in the uh in the uh in the uh in the uh uh more traditional readings of predictive processing make uh make clear as in uh idea that you are inferring about the the source of stimuli and not in uh and not and this is not i mean this is more easily amenable to the idea that the brain or the cognitive system knows that the the world is so and so the environment is so and so instead of saying that the green system knows how to engage with the world and i believe the the the the point that an activists and uh ecological psychologists are trying to make is to uh in in also working on the the predictive coding and the free energy principle and etc is to show that these notions these frameworks could be amenable to the idea that the cognitive system is knowing how to engage it knows how to engage the environment no way it doesn't need to be read in a handbook but also uh philosophically you have a large at the beginning of the the last few decades uh a gigantic discussion about whether no cloud could be reduced to know that and then again you have a background of this another philosophical discussion about whether know how could be treated as an independent notion from you know that and whether you could um reduce one to the other and so forth so uh i think it doesn't stop there you cannot uh even though i can make a a claim for perception as a mother knowing how there's still the the background issue of whether know how could be treated as a know that this is something that i am also working on recently there's a lot to say there it's it's such an interesting idea especially with linguistic communication it's a little bit like the finger and the moon and talking about the moon but the finger is only where it is and we're not talking about the pointer we're talking about the referent and so to use a socio-cognitive form of communication in in cultured language or in written academic discourse to peer through that or to reimagine that and see even what seems to be like the pinnacle of know that and still see it as know how i guess the question there is how do we read papers yes of course yes i think it is uh it is uh uh it's yeah at the very least it seems to be to be a very big problem because if we try to to treat these cognitive capacities as you mentioned as a matter of know-how it seems that we still miss something about communication for instance and on the other hand i think and this is my intuition but i think that we could in principle talk about how uh understanding what you just said and trying to respond to it is a matter of knowing how to behave linguistically but i think that even this movement to treat high levels of cognition as ultimately a matter of know-of-know-how it still leaves something out i am at i mean personally i am uh displeased with it i think at some level we have got to say that well there is another kind of approach that is needed here to explain how we communicate for instance in very articulate man not to say that it does not involve and not know about but it's not just that i think that at least more than practical knowledge is needed

but maybe the key here is to think of practical knowledge as basic and necessary for other sorts of other other kinds of knowledge that are needed to engage in this sort of exchange

maybe maybe maybe the maybe this is one way i'm sure that well uh more has to be said about this to give them a more informative answer but this is a at least my current position is that know-how provides the basis of our engagement with the world and with others but that some juncture we have to to be able to develop other kinds of understanding to explain at least explain the complex sorts of uh exchanges we have in our daily lives such as reading papers as you said i i agree i think framing know that as the basis and then recovering know-how is a challenge and it's also a challenge to begin with the basal cognitive processes as being enacted as being enacted know-how's and to reassemble those know-how pieces into know-how

that again it's not necessarily even a simple challenge but would we rather confront the challenge of assembling

symbolic propositional declarative verbal etc cognitive processes from a more liminal ecological cognitive approach or try to recover the ecological from the turing machine in a sense so it seems like um you're speaking in support of grounding in ecological plausible basal cognitive processes and then working towards assembling the skyscraper

uh

yes yes yes yeah totally yeah i think it's well both uh uh those both both approaches are uh difficult to to assemble as you said to to to provide this uh sort of explanation either bottom up or top down but i am more favorable of course to the the bottom up approach i think it also for another reason it it seems to the point of departure seems to be a biological description of how we interact with the world and the other way around it seems to be much more difficult to explain uh how we come up with this uh biological questions such as our body morphology and such it seems that we have to first account for that and then we we get to the the other questions which are also hard to to answer but at the very least it i i i think it's uh the bottom up approach is more uh integrative you know it's uh the chances are uh it seems to be more uh in our favor in the sense and and the way you just kind of framed it made me think that if you were to program a computer from scratch it could be an abstract or a virtual computer so it doesn't even need to have a specific embodiment and in that situation one might be concerned with uh assembling know that assert that and assembling these declarative statements and logic processes into something that is more dynamical or in interacting with the environment however that process of like full stack engineering with logic and silicone at the base is quite different from how we got here and so it might but it's it's complimentary um well you gave a little visibility into uh i guess the before during and after of peer review so i was just curious what changed during and after the paper like over the last months or years or however long it was how did you evolve on this and then what do you see as relevant next steps whether they're for you or just general to be done are we looking for a brain measurement are we looking for a video camera in a forest are we looking for a conceptual advance what is going to move forward past your developments that's that's a great question i think um i i'm not completely satisfied with the way i i've been for in the paper at the conceptual level i don't know if this

any of this any of this would be carried out in uh i honestly don't know if you could in principle carry it out in other empirical terms but uh i'm not saying that uh well one is i would write it a bit differently today not that it's uh not that i disagree with the paper originally but i think i there's something some points that i would like to reframe but they do not authorize uh i i think yet a new paper uh maybe in the future but this is something that uh the questions such as the myth of given which is a stupid philosophical question and uh conceptual one they still uh bother me but i have yet to find the opportunity to develop these ideas more carefully and

uh the general view i put forth there i think i i think it still stands uh but um i would like i i still i'm waiting for this chance to elaborate it further and see how far i can get but i don't know yet i must admit i don't know what are the the future developments of this idea but um i've been working on several isolated issues on this approach for the last few years with the notion of information and scientific understanding and information for ecological and and activists of cognition and recently i am working more recently with a few colleagues on virtual reality and perception cognition so uh this is how i like to conduct my research to find interesting uh questions and see how how can i deal with them but not necessarily find myself uh tied to one one line of reasoning in my way so this is what happened at the time i was working with rationality i thought it was a very cool subject and uh i still think it is but i haven't had the time to come back and reframe what i wanted to but uh maybe maybe sometime in the near future i don't know but more than the most recent thing i've been working on does not involve in study rationality but i think some parallels could be drawn which is that of uh virtual reality and how does uh visual reality actually compares to combination because well there's obviously something missing

from a perceiving in a virtual environment right um there is of course if i someone for instance uh if i if i'm on a virtual environment where i am under a kind of threat i may feel something analogous to fear but i'm not really

i'm not really fearing for my life and uh immersion does that to the person on the virtual reality so is it rational to to run away from the the the threat that the virtual threat and so on so forth this is this the the this is what i'm working on and uh maybe it's it provides a point of return to this uh idea of radical inactive rationality and uh in fact uh the the paper just got accepted so as i was telling you before i was so busy these last few days because we were correcting proofs my colleagues and i was uh with a tight schedule to to submit the corrected version of the paper in the last few weeks and then i got the proofs and so well maybe uh in the future i can come back to the issue of rationality i'd like to but i don't know if i i don't know i i really can't foresee the developments of that but uh in general i still agree with my original idea but i'll frame it differently a little bit today

so the vr point i'm sure that there's a lot more to say there but it reminded me of just the title virtual reality it's a virtualization and so in the same way that cognitive action planning allows us to entertain policies that our body doesn't take if only to evaluate them and choose against them but we can go on a flight of fam's fancy have a whimsical policy selection process of counterfactuals and then our morphology is kind of like our lifeline you know if you actually fall off

a ledge then it's going to hurt and then in virtual reality your embodied perceptual experience is quite literally a virtualization of survival critical situations so you might see yourself as being on a ledge but you're on a flat floor and then in that situation if you're kind of um a cognitive observer outside of the video game you might say well it's rational for me just to fall off to see what happens so that's like but then there's a layer of rationality within the space yes like what's a good move in the game um and so it's kind of blurring the line between the space and um it's it's blurring the criticality of the body and um

um i'm sure you've thought about this much more so how does this play into our virtual reality um um with the ways we think about and use and design these systems well this is the the the the questions that motivated us to to write a paper uh nada given them and i was that in the technical literature you see that a lot of researchers working with vr treat uh virtual perception as illusion a kind of illusion but we were a bit bothered by that because on the other hand it only works in vr because we we embody the devices devices they work for us you have the controllers you have the the headset and uh it seems so weird to say that the user the the user is under a kind of illusion and on the other hand it seems that it's not actually cognition because as you said there is no there is no biological uh relevance to those uh cognitive states so to speak for instance if i see an apple in vr it's it does not afford nourishing but it makes sense within the game right but it's different uh well when i perceive an apple in a in an actual environment it is it affords nourishment it is something i can eat but not in vr i can pretend to so we coined this uh concept of illusion to explain uh it is as if we were facing an actual level but it involves also this volition of play this idea that we are not uh being misled because we put on the equipment we know we are in the game and this i think is a hot subject right now because there's metaphors and i am very skeptical of whatever that can people are very optimistic but about the possibilities of the metaverse and some authors really think that we could be in a simulation strictly speaking but i mean the the best technology the current technology shows us is that it's very limited and you can pretend to be in a certain environment and so forth and this i think uh may offer a new uh new ways of thinking of thinking about rationality in this visual environment in ways that maybe russian to do something in that environment as you said to jump off the ledge and see what happens and uh on the other hand behavior aspects of doing so sometimes people actually feel fear and people for instance tweet for phobias with virtual realities and um it's it's a lot of there's a lot of interesting questions there but maybe i mean for future research you can yeah back to it it made me think about how the apple in real life it could be calories for you but also it could be calories for a fly it could be thrown you could use it as a hammer you could paint it so ecology and the generative process as we'd say in active inference kind of is always going beyond whatever affordances were programmed in for a given object and then also um the current discourse on vr focuses heavily on headsets just speaking totally anecdotally not not not my expert area um it seems a little bit interesting it's like as if the mobile phone discussion was only focused on rectangles of a certain size what makes this not virtual reality it's like looking through a window and there's another fellow to speak with so it's not attached to

my head it doesn't have some of the motion tracking um capacities that some technologies have but it is a niche modification in peripersonal space and we can push signals and affordances to each other so having a vr framing or really just a framing of rationality and perception cognition action cycles that aren't just constrained to one specific realization or hardware type and could run the gamut from total offline

person in a field to somebody who's very in the matrix seems pretty relevant otherwise new technologies even

little variants and software updates will continue to just infinitely surprise and confuse

yeah i think it's a it's a it's hard to define the the boundaries of reality in i mean we are here

uh in an exchange which i think it could in principle be classified as a form of virtual reality of course i'm

seeing you're in your own you see mine but uh i mean most i think works out they nowadays they they

define it a doc by saying that well must have a headset and sometimes a controller but but other than

that it's not virtual reality but yeah it's hard to define i think uh it's not not it's not a not as

straightforward as one would think because on uh talking commercially of course virtual reality means

headsets and and other paraphernalia but but a phone just talking on the phone or a telegraph in that

time it would have been on that interface with that sensory modality and with those affordances

it is the virtualization yeah that we have access to of just a different chapter today yeah i think so

yes part of our paper was to argue that in a way embodying technological devices is obviously not that's

not new it is the mark of being human it's embodying technological advice that what makes us what are we

are since the beginning using lithic technology and the fact that we can embody it so proficiently

a device even if it is programmed to for instance uh in within virtual reality to have something

a creature you control a creature with uh new morphology another morphology for instance even

if that's the case it only happens so proficiently because we are able to embody the device some

experiments have shown that you can map for instance a twist of your wrist to a third arm

that in virtual reality goes from was that from your chest and people can learn to control this

third arm that was mapped into an actual morphology but it can be made to solve by a twist of the wrist

so this is something we discussed in our paper that well you can you can only do that because

you embody the device so you can see it's an illusion it's actually well part of what means being human is

to embody technological devices so i would go that way i mean i know it blurs the distinction between

virtual and virtual but i think that's one way of dealing with this

cool is there anything else you want to talk about ask or just kind of prelude

well i'd like to thank you for the opportunity to discuss this paper it was very nice for me i i really

enjoyed it and this is a very nice and uh comfortable environment to discuss these issues and thank thank you

once again thank you yeah awesome discussion and important topics and we'll be happy to

talk to you again anytime okay thank you farewell thank you

thank you

Thank you.